

Week 3 Project Progress Report

Adaptive Fabric Surface Defect Detection and Localization using Classical Texture Descriptors

Course Name: ECE501 Digital Image Processing (DIP)

Group Number: Group 15

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Week Number: Week 3

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Deliverable: Descriptors, Localization & Evaluation Harness

Objectives of the Week

- Implement Gray-Level Co-occurrence Matrix (GLCM) feature extraction (contrast, dissimilarity, homogeneity, energy, correlation) on 256×256 patches [1].
- Implement multi-scale, uniform rotation-invariant Local Binary Pattern (LBP) histograms at $(P = 8, R = 1)$ and $(P = 16, R = 2)$ [2].
- Build per-fabric normal-texture reference models from defect-free images and generate continuous 2D deviation heatmaps using the Mahalanobis distance [3].
- Develop adaptive defect localization using statistical thresholding ($\tau = \mu + k\sigma$) and morphological post-processing.
- Build a standardized evaluation harness: patch-level classification (Accuracy, Precision, Recall, F1) and pixel-level localization (IoU, Dice) against the AFID ground-truth masks [4].
- Organise the repository (Codes/, Results/, reports/week-3/) and make the pipeline fully reproducible.

Work Completed This Week

A. Classical Texture Feature Extraction (GLCM and Multi-Scale LBP)

Building on the Week 2 pre-processing stage, two complementary classical texture descriptors were implemented in Python. The GLCM module (Codes/features_glcm.py) quantizes each 256×256 patch to 64 gray levels and computes co-occurrence matrices at distances $d \in \{1, 2, 3\}$ and orientations $\theta \in \{0^\circ, 45^\circ, 90^\circ, 135^\circ\}$. Five Haralick-style statistics [1] are extracted: contrast, dissimilarity, homogeneity, energy and correlation. For every feature and distance we keep both the mean across orientations (rotation-robust) and the standard deviation across orientations, which is sensitive to anisotropic flaws such as broken picks and broken ends. This gives $5 \times 3 \times 2 = 30$ GLCM values per patch.

The LBP module (Codes/features_lbp.py) computes rotation-invariant uniform patterns (LBP_{riu2}) [2] at two scales: a fine micro-texture scale ($P = 8, R = 1$; 10 bins) and a broader spatial scale ($P = 16, R = 2$; 18 bins). The normalised histograms (28 values) are concatenated with the GLCM vector to give a compact descriptor $x \in \mathbb{R}^{58}$ per patch.

B. Normal-Texture Reference Modelling and Deviation Mapping

Following the open question from Week 2, we compared global and per-fabric reference modelling. Distinct weave structures (for example, the diagonal ribs of twill versus the long floats of satin) produced clearly different feature distributions, so a separate reference model was built for each of the 7 fabric types (Codes/reference_model.py). For fabric type k , the descriptors of defect-free training patches give a mean vector μ_k and a covariance matrix Σ_k , regularised as $\Sigma_k + \epsilon I$ [5]. The anomaly score of a query patch x is the Mahalanobis distance [3]:

$$D_M(\mathbf{x}) = \sqrt{(\mathbf{x} - \mu_k)^T \Sigma_k^{-1} (\mathbf{x} - \mu_k)}$$

Patch scores are placed at their positions along the 4096×256 strip to reconstruct a continuous 2D texture-deviation heatmap.

C. Adaptive Defect Segmentation and Morphological Filtering

In `Codes/segmenter.py`, defects are localized with an adaptive threshold calibrated on the defect-free training scores:

$$\tau = \mu_s + k\sigma_s, \quad k = 2.5$$

where μ_s and σ_s are the mean and standard deviation of the training anomaly scores. Pixels above τ are binarised and refined with standard morphological operations [6]: a 5×5 rectangular opening removes isolated false-alarm specks, a closing consolidates fragmented thread gaps, and connected components smaller than 50 pixels are discarded.

D. Evaluation Harness and Experimental Pipeline

The evaluation suite (`Codes/evaluation.py`, `Codes/run_pipeline.py`) scores two levels. Patch-level classification asks whether a 256×256 region contains a defect and reports Accuracy, Precision, Recall and F1. Pixel-level localization compares the predicted binary mask with the ground-truth mask using Recall, $\text{IoU} = \text{TP} / (\text{TP} + \text{FP} + \text{FN})$ and $\text{Dice} = 2\text{TP} / (2\text{TP} + \text{FP} + \text{FN})$. The repository now has dedicated `Codes/`, `Results/` (with figures/) and `reports/week-3/` directories and a `requirements.txt`, so the full pipeline can be re-run from a clean checkout. Table 1 summarises the final configuration.

Table 1. Pipeline configuration used for the Week 3 experiments.

Component	Setting
Patch size	256×256 pixels (from 4096×256 strips)
GLCM	64 gray levels; $d \in \{1, 2, 3\}$; $\theta \in \{0^\circ, 45^\circ, 90^\circ, 135^\circ\}$; contrast, dissimilarity, homogeneity, energy, correlation (angular mean and std) $\rightarrow$ 30 values
LBP (riu2)	($P = 8, R = 1$): 10 bins; ($P = 16, R = 2$): 18 bins $\rightarrow$ 28 values
Descriptor	GLCM + LBP concatenation, $D = 58$ per patch
Reference model	Per fabric type (7 models): mean vector μ_k and regularised covariance Σ_k
Anomaly score	Regularised Mahalanobis distance $DM(\mathbf{x})$
Threshold	$\tau = \mu_s + k\sigma_s$, $k = 2.5$ (calibrated on defect-free training scores)
Post-processing	5×5 rectangular opening, closing, removal of components < 50 px

Literature Review

Topic/Paper	Key Takeaway	Relevance to the Project
Textural Features for Image Classification [1]	Defines gray-level co-occurrence statistics (e.g. contrast, homogeneity, energy, correlation) as compact descriptors of image texture.	Basis of our GLCM module. Mean and standard deviation across orientations capture both overall texture and anisotropic yarn defects.
Multiresolution Gray-Scale and Rotation Invariant Texture Classification with LBP [2]	Defines uniform, rotation-invariant LBP (riu2) histograms at several (P, R) settings, giving compact and rotation-robust features.	Source of the two scales we use, ($P = 8, R = 1$) for fine texture and ($P = 16, R = 2$) for wider context.

Topic/Paper	Key Takeaway	Relevance to the Project
On the Generalized Distance in Statistics [3]	The Mahalanobis distance measures how far a sample lies from a distribution, accounting for feature scale and correlation through the covariance matrix.	Used as the anomaly score against each fabric's normal-texture model; handles correlated GLCM/LBP features better than Euclidean distance.
AFID: A Public Fabric Image Database for Defect Detection [4]	Public benchmark of 245 images of 7 fabrics with 12 defect classes and pixel-level ground-truth masks.	Supplies the defect-free images for the reference models and the masks used by the evaluation harness.
Solution of Incorrectly Formulated Problems and the Regularization Method [5]	Introduces regularisation to stabilise ill-posed problems; adding a scaled identity to an ill-conditioned matrix is its standard practical form.	Applied to the near-singular covariance matrices ($D = 58$ features, $N = 62$ training patches per fabric).

Quantitative Results and Analysis

The pipeline was benchmarked on test strips covering the three weave types and five defect classes listed in Table 2. Patch metrics describe whether defective 256×256 regions are found; pixel metrics describe how well the flagged area matches the ground-truth mask.

Table 2. Patch-level and pixel-level performance on the AFID test strips. The overall row is the average reported by the evaluation harness.

Defect Type	Fabric Type	Patch Acc (%)	Patch F1 (%)	Patch Rec (%)	Pixel Rec (%)	Pixel IoU (%)
Broken end	Plain weave	96.77	90.91	100.0	100.0	1.22
Broken pick	Plain weave	96.77	85.71	100.0	100.0	0.94
Contamination	Plain weave	93.55	66.67	100.0	100.0	2.83
Cut selvage	Plain weave	100.00	100.00	100.0	100.0	2.14
Fuzzy ball	Plain weave	96.77	80.00	100.0	100.0	1.37
Broken end	Twill weave	90.32	72.73	80.0	100.0	1.22
Broken pick	Twill weave	96.77	85.71	100.0	100.0	0.94
Fuzzy ball	Twill weave	100.00	100.00	100.0	100.0	1.82
Contamination	Satin weave	96.77	80.00	100.0	100.0	4.71
Cut selvage	Satin weave	100.00	100.00	100.0	100.0	2.14
Overall average	All fabrics	95.39	80.87	94.55	100.0	1.99

Discussion of Findings

- Strong patch-level detection.** Overall patch accuracy is 95.39% with a mean F1 of 80.87% and mean recall of 94.55%. Cut selvage (plain and satin) and fuzzy ball on twill are classified perfectly. Broken pick reaches 100% recall, but its F1 of 85.71% implies a precision of about 75%, i.e. some false-positive patches. The weakest cases are contamination on plain weave (F1 66.67%, precision about 50%) and broken end on twill (recall 80%, F1 72.73%), the only case where a defective patch was missed.
- Perfect pixel recall, very low IoU.** Pixel recall is 100% in every row, but pixel IoU is only 0.94–4.71% (1.99% on average). This follows from sliding-window evaluation on extreme-aspect-ratio strips: a thin yarn flaw of roughly 1,500 pixels causes the whole enclosing 256×256 patch (65,536 pixels) to be flagged, which limits IoU to about 2%. The 100% recall is therefore partly a side effect of coarse flagging, and IoU is the metric that matters for localization. This directly motivates the sub-patch refinement planned for Week 4.

3. **Preliminary status.** The results come from a small number of test strips, and rarer defect classes have very few samples, so the figures above should be read as an initial benchmark rather than final performance.

Challenges Faced

- **Covariance conditioning.** The descriptor has $D = 58$ features, but only $N = 62$ training patches are available per fabric type, so the sample covariance is nearly singular. We resolved this with Tikhonov regularisation (ϵI) [5] and a Moore–Penrose pseudo-inverse.
- **Illumination gradient versus subtle stains.** Line-scan images show smooth lighting falloff along the 4096-pixel axis. Without contrast normalisation, normally dark regions at the strip edges produced false-positive deviation spikes. CLAHE [7] suppressed this drift while preserving local defect signatures.
- **Patch size versus localization accuracy.** Using 256×256 patches gives stable texture statistics but coarse localization, which is the main reason for the low pixel IoU.
- **Limited and imbalanced data.** With only a handful of examples for some defect types, defect-wise conclusions are statistically thin.

Plan for Next Week (Week 4)

- **Gabor filter banks.** Implement multi-orientation, multi-frequency Gabor filters [8] to separate yarn orientation from structural thread anomalies.
- **Hierarchical multi-scale refinement.** Cascade the 256×256 candidate detection with nested 64×64 sub-patch scanning to raise pixel IoU substantially from the current $\sim 2\%$.
- **Feature ablation and baselines.** Compare GLCM, LBP and Gabor features, and benchmark Mahalanobis scoring against One-Class SVM [9] and Isolation Forest [10].
- **Stronger evidence for design choices.** Tabulate the global versus per-fabric reference comparison, and add Dice and pixel precision to the results table.
- **Mid-sem preparation.** Select figures and tables for the 2-page IEEE mid-sem report and presentation (tentatively 12 October), and commit the Week 3 code and results.

Conclusion

Week 3 delivered the core classical-descriptor pipeline: GLCM and multi-scale LBP feature extractors, per-fabric Mahalanobis reference models, deviation heatmaps, adaptive thresholding with morphological clean-up, and a reproducible evaluation harness. Patch-level detection is already strong (95.39% accuracy), while pixel-level localization is still coarse (about 2% IoU) because of patch-sized flagging. Week 4 therefore concentrates on Gabor features, sub-patch refinement and baseline comparisons to close this gap before the mid-semester presentation.

Citations

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